

Time Series Analysis

Chapter 3: Unit roots and ARIMA models



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CUSUM**: CUmulative SUM (control chart)
- **DF**: Dickey–Fuller (test)
- **DF-GLS**: Dickey–Fuller test with GLS detrending
- **DS**: Difference-Stationary (series)
- **DW**: Durbin–Watson (statistic)
- **EU**: European Union
- **FRED**: Federal Reserve Economic Data

Acronyms used in this material (2/2)

- **GDP**: Gross Domestic Product
- **GLS**: Generalised Least Squares
- **GNP**: Gross National Product
- **HICP**: Harmonised Index of Consumer Prices
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **MA**: Moving Average
- **MAIC**: Modified Akaike Information Criterion
- **OLS**: Ordinary Least Squares
- **PACF**: Partial AutoCorrelation Function
- **PP**: Phillips–Perron (test)
- **RMSE**: Root Mean Squared Error
- **SARIMA**: Seasonal ARIMA (model)
- **SE**: Standard Error
- **TBATS**: Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)
- **TS**: Trend-Stationary (series)
- **US**: United States
- **VAR**: Vector AutoRegression
- **ZA**: Zivot–Andrews (test)

Today's question and route

- **Question**: when a series drifts upwards, is the shock of today temporary or permanent, and how should we forecast it?
  - the answer decides whether we detrend or difference, and how wide our forecast intervals are
- **Route** of the chapter
  - deterministic and stochastic trends; integrated processes $I(d)$; spurious regression
  - unit-root tests: Dickey–Fuller, ADF, Phillips–Perron; the KPSS stationarity test; structural breaks
  - ARIMA(p, d, q) models: identification, estimation, diagnostics, forecasts with growing intervals
- We build on Chapter 1 (stationarity, random walk, differencing) and Chapter 2 (ARMA models); Seminar 3 comes before this lecture

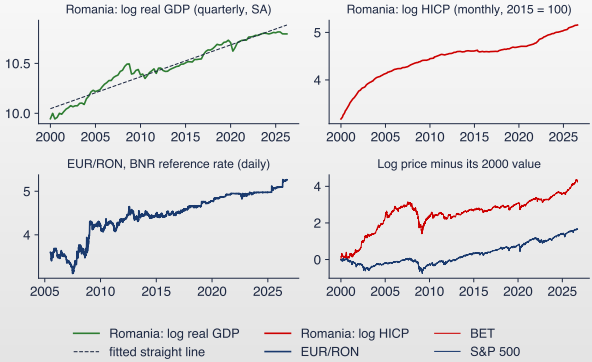
Learning outcomes

- Distinguish trend-stationary from difference-stationary series, and explain why the distinction matters
- Recognise a spurious regression and explain why it happens
- Run and interpret the ADF, Phillips–Perron and KPSS tests, with the right deterministic terms and lags
- Explain how a structural break can mimic a unit root
- Identify, estimate, check and forecast an ARIMA(p, d, q) model, and explain why its intervals grow with the horizon
- Use automatic ARIMA selection critically

Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 4–5 (ARIMA models and non-stationarity)
  - companion, free online: Hyndman and Athanasopoulos (2021), Ch. 9 (ARIMA models)
- Theory: Hamilton (1994), Ch. 15–17 (trends, unit roots); Box, Jenkins and Reinsel (2008) for the ARIMA methodology
- Python Quantlets of this chapter: Quantlets/Ch_03
  - `adfuller`, `kpss`, `zivot_andrews` and `ARIMA` from `statsmodels`; the Phillips–Perron test written out in a few lines
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Four series that drift



- Romanian real GDP (Eurostat, seasonally adjusted) and HICP (Eurostat); EUR/RON, the BNR reference rate; BET and S&P 500 log prices, daily since 2000
- None of them has a constant mean: Chapter 1 called them non-stationary

TSA_ch3_trends

Interpreting the four series

- GDP grows by 0.80% per quarter on average (about 3.2% per year), but it strays from the straight line by up to 18%
  - the 2009 recession: did GDP come back to the old line, or did it start a new one?
- Consumer prices: $\times 7.2$ since 2000; EUR/RON: from 3.60 to 5.26 lei; BET: $+426$ log points, S&P 500: $+166$ log points (to 18 September 2026)
- Two models can produce such pictures
  - a fixed line plus stationary noise (**deterministic trend**)
  - a random walk with drift (**stochastic trend**)

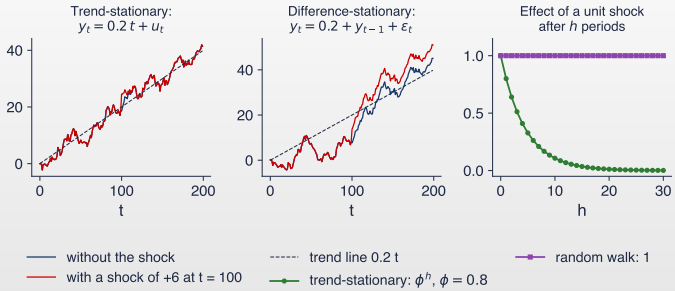
Trend-stationary series

- **Definition**: $y_t = \alpha + \beta t + u_t$, with u_t a stationary process (for example a stationary AR(1), Chapter 2)
  - α: the intercept; β: the slope, the average change per period; t: the time index
  - y_t is **trend-stationary** (TS): stationary after subtracting the line $\alpha + \beta t$
- Properties
  - $E[y_t] = \alpha + \beta t$; $\text{Var}(y_t) = \text{Var}(u_t)$, constant
  - a shock moves y_t away from the line only temporarily: with $u_t = \phi u_{t-1} + \varepsilon_t$, its effect after h periods is ϕ^h
  - long-run forecasts return to the line; their intervals have a bounded width
- Right treatment: estimate the trend by OLS (ordinary least squares) and model the residuals as ARMA

Difference-stationary series

- **Definition**: $y_t = \beta + y_{t-1} + u_t$, u_t stationary; by substitution, $y_t = y_0 + \beta t + \sum_{i=1}^{t} u_i$
  - β: the drift, the average change per period; y_0: the starting value
  - y_t is **difference-stationary** (DS): $\Delta y_t = \beta + u_t$ is stationary
  - $\sum_{i \le t} u_i$ is the **stochastic trend**; βt is the drift
- Properties
  - the same mean $y_0 + \beta t$ as a TS series, but $\mathrm{Var}(y_t)$ grows with t (Chapter 1: $t\sigma^2$ for a random walk)
  - every shock stays in the level forever: there is no line to return to
  - forecast intervals widen without limit as the horizon grows
- Right treatment: difference the series and model Δy_t as ARMA: the ARIMA model of this chapter

Trend-stationary and difference-stationary series: the effect of a shock



- Left and centre: the same shocks drive a TS series (u_t AR(1), $\phi = 0.8$) and a random walk with drift 0.2; the red paths add one shock of $+6$ at $t = 100$
- Right: the effect of a unit shock after h periods

TSA_ch3_trends

Interpreting the two trends

- TS series: the shock dies out; its effect is $0.8^{10} = 0.107$ after 10 periods and 0.012 after 20
  - at $t = 200$ the two red and blue paths coincide
- Random walk: the whole path shifts up by 6 and stays there
  - a recession in a DS economy is a permanent loss of output, not a dip below a trend
- By eye, the two series without the shock look alike: we need a formal test (Sections 3–4)

Integrated processes

- **Definition**: y_t is **integrated of order** d, $y_t \sim I(d)$, if $\Delta^d y_t$ is stationary and $\Delta^{d-1} y_t$ is not
  - $I(0)$: stationary (with an invertible ARMA representation, Chapter 2); $I(1)$: stationary after one difference; $I(2)$: after two
  - $\Delta^2 y_t = \Delta y_t - \Delta y_{t-1} = y_t - 2y_{t-1} + y_{t-2}$
- Typical orders in economics and finance
  - log prices of stocks, exchange rates, log GDP: $I(1)$; returns and growth rates: $I(0)$
  - price levels in high-inflation periods: possibly $I(2)$, so that inflation itself is $I(1)$
- Rules
  - $I(0) + I(1) = I(1)$: the stochastic trend dominates; a TS series is $I(0)$ after removing the trend, not after differencing

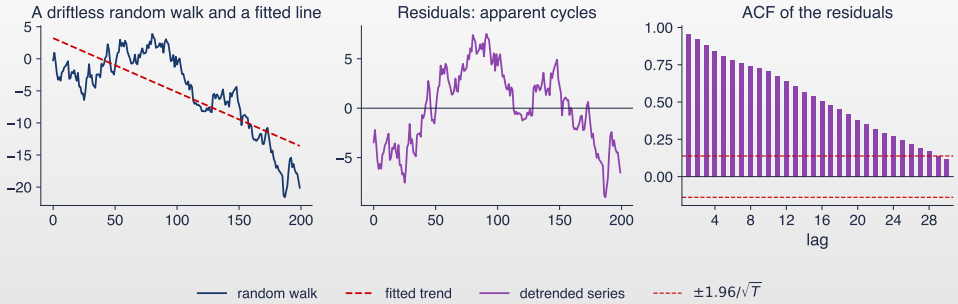
The unit root

- Write an AR(p) with the lag operator L (Chapter 1): $\phi(L)y_t = \varepsilon_t$, $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$
  - Chapter 2: stationary if all roots of $\phi(z) = 0$ lie outside the unit circle, $|z| > 1$
- **Unit root**: one root equals $z = 1$, so $\phi(z) = (1 - z)\,\phi^*(z)$ with the roots of ϕ^* outside the circle
  - then $\phi^*(L)\,\Delta y_t = \varepsilon_t$: Δy_t is a stationary AR($p - 1$), and $y_t \sim I(1)$
  - AR(1): $\phi(z) = 1 - \phi z$ has the root $1/\phi$, a unit root when $\phi = 1$: the random walk
- **Example**: $y_t = 1.5y_{t-1} - 0.5y_{t-2} + \varepsilon_t$; $1 - 1.5z + 0.5z^2 = (1 - z)(1 - 0.5z)$
  - roots 1 and 2: $\Delta y_t = 0.5\,\Delta y_{t-1} + \varepsilon_t$, an AR(1) in differences

Case study: Nelson and Plosser (1982)

- **Question**: are US macroeconomic series stationary around a trend, as the business-cycle models of the 1970s assumed?
  - Nelson and Plosser (1982): 14 long annual series (real GNP, employment, prices, wages, interest rates, stock prices), some since 1860
- **Method**: the Dickey–Fuller test of Section 3, with a constant and a trend
  - the unit root was rejected for one series only, the unemployment rate
- **Consequence**: shocks to output look permanent, not transitory
  - a large debate on the nature of recessions; and a warning against detrending by a straight line (next slide)
  - the answer was later challenged by structural breaks (Section 5)

Detrending a random walk



- Left: a driftless random walk ($T = 200$) and the OLS line $\hat{a} + \hat{b}t$; centre: the residuals; right: their ACF

TSA_ch3_trends

Interpreting the detrended random walk

- The line fits: $R^2 = 0.61$, $t = -17.7$ for the slope, although the true slope is 0
  - over 2 000 random walks: $|t| > 1.96$ in 90% of cases, median $R^2 = 0.44$
- The residuals look like long cycles: $\hat{\rho}(1) = 0.96$ and a slow decay
  - Nelson and Kang (1981): detrending a DS series creates **spurious cycles**; a business cycle may be an artefact of the method
- The reverse error (differencing a TS series) is over-differencing (Chapter 1 and Section 6)

Recap: Deterministic and stochastic trends

- TS: $y_t = \alpha + \beta t + u_t$, shocks fade; DS: $\Delta y_t = \beta + u_t$, shocks are permanent
- $I(d)$: stationary after d differences; a unit root in $\phi(z)$ means $I(1)$
- The wrong treatment creates artefacts: spurious cycles (detrending a DS series) or over-differencing (differencing a TS series)

1926: nonsense correlations

- Yule (1926) asked: *Why do we sometimes get nonsense-correlations between time-series?*
  - his example: in England and Wales, 1866–1911, the share of Church of England marriages and the mortality rate had a correlation of about 0.95
  - no causal link: both series simply declined over the period
- His explanation: for series that wander like sums of random terms, the sample correlation does not settle near 0
  - large values occur often by chance; the usual standard error does not apply
- Fifty years later, the same problem reappeared in econometric models estimated on levels

1974: Granger and Newbold

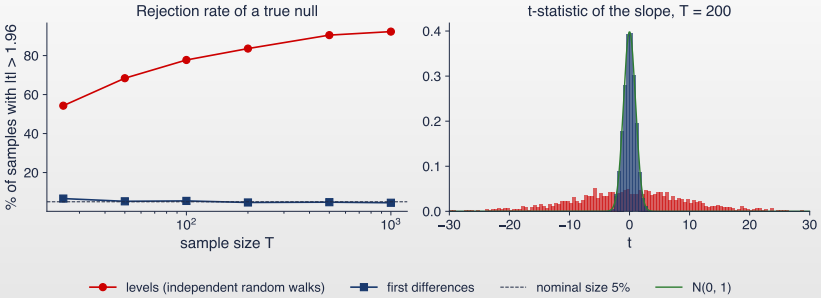


Clive Granger (1934–2009), Nobel Prize in Economics 2003

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

- Granger and Newbold (1974): regress one random walk on another, **independent** random walk
  - $y_t = a + bx_t + e_t$ with y_t, x_t independent $I(1)$ series
  - the t-test of $b = 0$ rejects far too often; R^2 is high; the Durbin–Watson statistic is close to 0
- Their rule of thumb: be suspicious when $R^2 >$ DW
  - DW (Durbin–Watson) $\approx 2(1 - \hat{\rho}_1)$ of the residuals: near 0 means $I(1)$-like residuals
- Granger and Newbold were then at the University of Nottingham; Granger later received the Nobel Prize for cointegration (Chapter 7)

Spurious regression by simulation



- 2 000 pairs of independent Gaussian random walks for each T; OLS of y on a constant and x, in levels and in first differences
- Left: share of samples with $|t| > 1.96$; right: the distribution of t for $T = 200$

TSA_ch3_spurious_regression

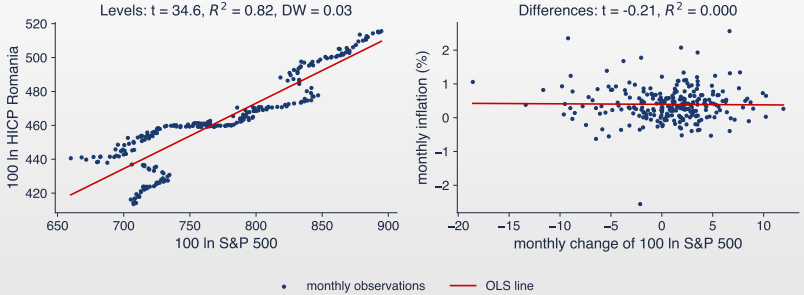
Interpreting the simulation

- Levels: a true null $b = 0$ is rejected in 68% of samples with $T = 50$ and in 92% with $T = 1000$
  - more data makes it worse: the median $|t|$ grows from 3.2 ($T = 50$) to 14.2 ($T = 1000$)
  - median R^2 about 0.17 for every T; median DW 0.08 at $T = 200$
- Differences: 4.7% at $T = 200$, close to the nominal 5%
  - the t-test works again because Δy_t and Δx_t are stationary

Why the t-test fails

- Under $b = 0$ the error $e_t = y_t - a$ is itself a random walk: the OLS assumptions (stationary, uncorrelated errors) fail
  - the usual standard error is far too small
- Phillips (1986) derived the large-sample behaviour
  - $\hat{b}$ does not converge to 0: it converges to a random variable
  - $t_{\hat{b}}$ grows like $\sqrt{T}$, so the rejection rate tends to 100%
  - R^2 has a non-degenerate limit distribution; DW $\to 0$
- Exception: if $y_t - bx_t$ is stationary for some b, the series are **cointegrated** and the levels regression is meaningful Engle and Granger (1987) (Chapter 7)

A spurious regression on real data



- Romanian HICP (Eurostat) against the S&P 500 (month-end close), 260 months since January 2005, both as 100 ln
- Left: levels; right: monthly changes

TSA_ch3_spurious_regression

Interpreting the HICP regression

- Levels: slope 0.39, $t = 34.6$, $R^2 = 0.82$, DW $= 0.03$
  - "US stocks explain Romanian consumer prices": $R^2 >$ DW, a textbook spurious regression
- Differences: $t = -0.21$ (p $= 0.83$), $R^2 = 0.0002$
  - monthly inflation and monthly stock returns are unrelated
- Both levels contain a trend; the regression only says that both went up

Recap: Spurious regression

- Regressing one $I(1)$ series on an unrelated $I(1)$ series gives large t, high R^2 and DW near 0
- More observations do not help: t grows like $\sqrt{T}$
- Remedies: test the order of integration first; regress differences; or test for cointegration (Chapter 7)

Testing $\phi = 1$

- AR(1): $y_t = \phi y_{t-1} + \varepsilon_t$; subtract y_{t-1}: $\Delta y_t = \gamma y_{t-1} + \varepsilon_t$, with $\gamma = \phi - 1$
  - H_0: $\gamma = 0$ (unit root, $I(1)$); H_1: $\gamma < 0$ (stationary); a one-sided, left-tail test
  - statistic: $\tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$, the usual OLS t-ratio; SE: the standard error; a very negative τ speaks against the unit root
- Under H_0, τ is **not** Student or Normal Dickey and Fuller (1979)
  - y_{t-1} is a random walk, so the regressor is not stationary; $\hat{\phi}$ converges at rate T, not $\sqrt{T}$ (**superconsistency**)
  - the limit of τ is a functional of Brownian motion, skewed to the left Hamilton (1994), Ch. 17
  - critical values come from simulation: Fuller (1996); today from response surfaces MacKinnon (2010)

Iowa State: where the test was born



Snedecor Hall, Iowa State University: the department of statistics and its Statistical Laboratory (1933)

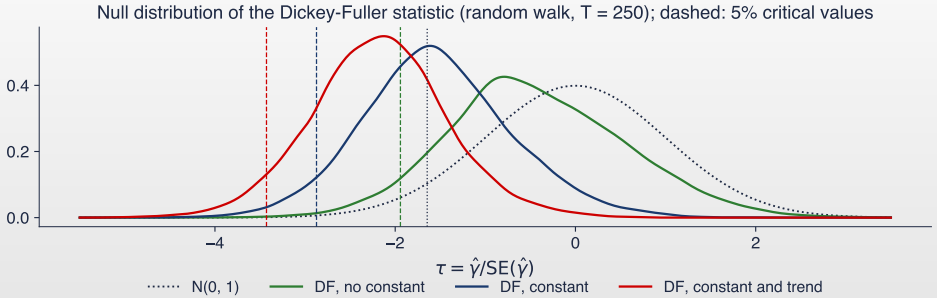
Photo: Maitra (2023); CC BY-SA 4.0; Wikimedia Commons

- **Wayne A. Fuller**: statistician at Iowa State, author of *Introduction to Statistical Time Series* Fuller (1996)
- **David A. Dickey**: his doctoral student at Iowa State (1976), later at North Carolina State University
- Dickey and Fuller (1979): the distribution of $\hat{\phi}$ and τ under a unit root; Dickey and Fuller (1981): joint tests of the unit root and the deterministic terms
- Said and Dickey (1984): the augmented test for ARMA errors (this section)

Three Dickey–Fuller regressions

- **No constant**: $\Delta y_t = \gamma y_{t-1} + \varepsilon_t$
  - H_1: stationary with mean 0; rarely realistic
- **Constant**: $\Delta y_t = c + \gamma y_{t-1} + \varepsilon_t$
  - H_0: random walk (with drift if $c \neq 0$); H_1: stationary around a non-zero mean
  - for series without a trend: interest rates, inflation, exchange rates, returns
- **Constant and trend**: $\Delta y_t = c + bt + \gamma y_{t-1} + \varepsilon_t$
  - c: the constant; bt: a linear trend with slope b
  - H_0: random walk with drift; H_1: trend-stationary: exactly the TS against DS question of Section 1
  - for trending series: log GDP, log prices, log price indices
- Each case has its own distribution of τ and its own critical values

The Dickey–Fuller distributions



- 20 000 random walks with $T = 250$: the density of τ in the three regressions, against $N(0, 1)$; dashed lines: 5% critical values MacKinnon (2010)

TSA_ch3_dickey_fuller

Interpreting the three distributions

- All three lie to the left of $N(0,1)$; the more deterministic terms, the further left
  - simulated 5% quantiles: -1.95 (no constant), -2.86 (constant) and -3.42 (constant and trend); tabulated: -1.94, -2.87 and -3.43
- With the Normal critical value -1.645, a true unit root is "rejected" in 46% of samples (constant) and 77% (constant and trend)
  - the wrong table turns random walks into "stationary" series

Critical values of the Dickey–Fuller test

	no constant			constant			constant and trend		
T	1%	5%	10%	1%	5%	10%	1%	5%	10%
50	−2.61	−1.95	−1.61	−3.57	−2.92	−2.60	−4.15	−3.50	−3.18
100	−2.59	−1.94	−1.61	−3.50	−2.89	−2.58	−4.05	−3.46	−3.15
250	−2.57	−1.94	−1.62	−3.46	−2.87	−2.57	−4.00	−3.43	−3.14
∞	−2.57	−1.94	−1.62	−3.43	−2.86	−2.57	−3.96	−3.41	−3.13

- Source: the response surfaces of MacKinnon (2010), as used by `statsmodels`; p-values from MacKinnon (1994)
- Decision: reject the unit root at 5% if τ is **below** (more negative than) the 5% value
- The values depend only weakly on T; they apply unchanged to the ADF test

Worked example: Romanian real GDP

- $y_t = 100 \ln$ real GDP, 2000Q1–2026Q2; regression with constant and trend, no lags ($T = 105$ differences)
  - OLS: $\hat{\gamma} = -0.0961$, SE($\hat{\gamma}$) $= 0.0416$, so $\hat{\phi} = 1 + \hat{\gamma} = 0.904$
- $\tau = -0.0961/0.0416 = -2.31$; 5% critical value with constant and trend: -3.45
  - $-2.31 > -3.45$: do not reject the unit root (p $= 0.43$)
  - with the Normal value -1.645 we would have rejected it: the wrong table gives the wrong answer
- Reading: "not rejected" is not "proved"
  - if $\phi = 0.904$ were true, a shock would halve in $\ln 0.5 / \ln 0.904 \approx 6.9$ quarters; with 106 observations the test can hardly tell this from $\phi = 1$

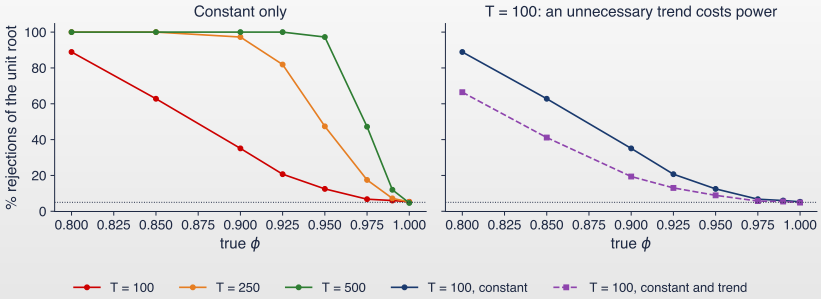
The augmented Dickey–Fuller test

- If Δy_t is autocorrelated, ε_t in the DF regression is not white noise and the size of the test is wrong
  - the remedy: add lagged differences
- **ADF** (augmented Dickey–Fuller) regression:
  - $\Delta y_t = c + bt + \gamma y_{t-1} + \sum_{j=1}^{k} \delta_j \Delta y_{t-j} + \varepsilon_t$; same τ, same critical values
  - δ_j: the coefficients of the past differences; k: the number of lagged differences
  - an AR(p) in levels becomes exactly this regression with $k = p - 1$ and $\gamma = -\phi(1)$
- Said and Dickey (1984): with ARMA errors the test stays valid if k grows slowly with T
  - the lags approximate the MA part by a long AR
- The ADF test is the default unit-root test in software: `adfuller` in `statsmodels`

Choosing the number of lags k

- Maximum: $k_{\max} = 12\,(T/100)^{1/4}$ Schwert (1989); for $T = 105$, $k_{\max} = 13$
  - then choose $k \le k_{\max}$ by an information criterion, AIC or BIC (Chapter 2), on the same sample for every k
- Too few lags: autocorrelated errors, wrong size (with a negative MA part, far too many rejections)
  - too many lags: lower power
- Ng and Perron (2001): a modified AIC (MAIC) gives a better size; GLS detrending Elliott, Rothenberg and Stock (1996) gives more power
  - MAIC: modified Akaike information criterion; GLS: generalised least squares; both appear in the DF-GLS test
- Check the result: Ljung–Box on the ADF residuals Ljung and Box (1978); report k together with τ

Size and power of the Dickey–Fuller test



- 2 000 AR(1) paths $y_t = \phi y_{t-1} + \varepsilon_t$ for each ϕ; share of rejections at 5%; at $\phi = 1$ this is the size

TSA_ch3_dickey_fuller

Interpreting the power curves

- Size: 5.30%, 5.20% and 4.65% at $\phi = 1$ (with a constant): the test holds its 5% level
- Power at $\phi = 0.95$: 12% with $T = 100$, 47% with $T = 250$, 97% with $T = 500$
  - a quarterly series of 25 years ($T = 100$) almost never distinguishes $\phi = 0.95$ from a unit root
  - what matters is the time span, not the frequency: daily data over two years do not help
- An unnecessary trend costs power: at $T = 100$ and $\phi = 0.9$, 35% with a constant, 19% with constant and trend

Choosing the deterministic terms

- The alternative must be a plausible description of the data
  - trending series (log GDP, log prices): constant and trend
  - series that fluctuate around a level (interest rates, inflation, exchange rates, returns): constant
- Omitting a needed trend: a TS series looks like a unit root; the test almost never rejects
  - adding an unneeded trend: lower power (previous chart)
- Look at the plot first, then choose; Dickey and Fuller (1981) give joint F-type tests of the unit root and the trend

How many differences: $I(1)$ or $I(2)$?

- Dickey and Pantula (1987): test from the highest plausible order downwards
  - step 1: ADF on Δy_t; if it does not reject, y_t may be $I(2)$
  - step 2: if it rejects, ADF on y_t; reject: $I(0)$; do not reject: $I(1)$
- Why downwards: if $y_t \sim I(2)$, the test on y_t assumes at most one unit root and is not valid
- Example: Romanian consumer prices since 2005 (the table in Section 4)
  - $\ln P$: unit root; inflation π: ADF p $= 0.32$, KPSS 0.404; change of inflation: clearly stationary
  - so $\ln P$ is $I(1)$ or $I(2)$, depending on how persistent inflation is: Section 7 returns to this

Recap: The Dickey–Fuller test

- $\Delta y_t = c + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$; H_0: $\gamma = 0$; τ against Dickey–Fuller critical values
- Deterministic terms follow the plot; lags by AIC or BIC up to $12(T/100)^{1/4}$
- Low power near $\phi = 1$: not rejecting is weak evidence for a unit root
- Order of integration: test from the top (Δy_t first)

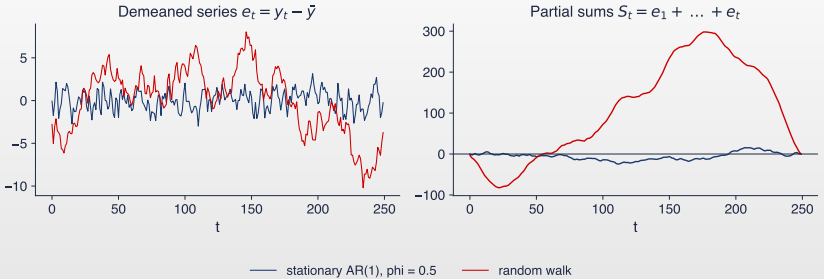
The Phillips–Perron test

- Phillips and Perron (1988): keep the simple DF regression (no lagged differences) and correct τ instead
  - $Z_\tau = \sqrt{\hat{\gamma}_0 / \hat{\lambda}^2}\,\tau - \dfrac{(\hat{\lambda}^2 - \hat{\gamma}_0)\, T\, \mathrm{SE}(\hat{\gamma})}{2\hat{\lambda}\, s}$
  - $\hat{\gamma}_0$: variance of the residuals; s^2: their OLS variance; $\hat{\lambda}^2$: their **long-run variance**
- Long-run variance (Newey–West, Bartlett weights): $\hat{\lambda}^2 = \hat{\gamma}_0 + 2\sum_{j=1}^{L}(1 - \frac{j}{L+1})\hat{\gamma}_j$
  - $\hat{\gamma}_j$: autocovariances of the residuals; L: the number of autocovariances included (the bandwidth)
  - if the residuals are white noise, $\hat{\lambda}^2 = \hat{\gamma}_0$ and $Z_\tau = \tau$: the correction only matters when they are autocorrelated
- Same H_0 and the same critical values as ADF; robust to heteroskedasticity; no lag choice, but a bandwidth L
  - weak point: large size distortions with a negative MA component Schwert (1989)

The KPSS test: stationarity as the null

- Kwiatkowski et al. (1992) (KPSS: Kwiatkowski–Phillips–Schmidt–Shin): $y_t = \xi t + r_t + \varepsilon_t$, $r_t = r_{t-1} + u_t$, $u_t \sim \text{WN}(0, \sigma_u^2)$
  - ξt: a deterministic trend; r_t: a random walk, the moving level; ε_t: stationary noise
  - H_0: $\sigma_u^2 = 0$ (the random walk is a constant: y_t stationary around a level or a trend); H_1: $\sigma_u^2 > 0$, a unit root
- Statistic: regress y_t on a constant (and a trend); residuals e_t; partial sums $S_t = e_1 + \cdots + e_t$
  - $\eta = \dfrac{1}{T^2 \hat{\lambda}^2} \sum_{t=1}^{T} S_t^2$, with $\hat{\lambda}^2$ the long-run variance of e_t
  - reject stationarity for **large** η: 5% critical values 0.463 (level) and 0.146 (trend); 1%: 0.739 and 0.216
- A test of stationarity, not of a unit root: it reverses the burden of proof

KPSS: why partial sums



- A stationary AR(1) ($\phi = 0.5$) and a random walk, $T = 250$: the demeaned series and their partial sums S_t
- KPSS (level): 0.16 for the AR(1), 0.73 for the random walk; 5% critical value 0.463

TSA_ch3_unit_root_tests

Interpreting the partial sums

- Stationary series: deviations cancel out, S_t stays close to 0
  - $\sum S_t^2$ grows like T^2, so η stays bounded
- Random walk: long runs on one side of the mean, S_t makes large excursions
  - $\sum S_t^2$ grows like T^4: η grows with T and the test rejects
- The same idea as the CUSUM charts of quality control: cumulated deviations reveal a drifting level

ADF and KPSS together

	KPSS does not reject	**KPSS rejects**
ADF rejects	$I(0)$: both agree	conflict: break, long memory, outliers?
ADF does not reject	inconclusive: the data cannot decide	$I(1)$: both agree

- The two tests have opposite null hypotheses: agreement is stronger evidence than either test alone
- Inconclusive is common with short samples: low power on both sides
  - then decide by the purpose: for forecasting, $d = 1$ is the cautious choice (wider intervals)
- Conflict: look for breaks (Section 5) or fractional integration (Chapter 8)

Unit-root tests on real series

Series	det.	T	ADF (p)	k	PP	KPSS	ADF+KPSS
Romania, real GDP, $100 \ln Y$	c, t	106	-2.31 (0.428)	0	-2.19	0.198	$I(1)$
its growth Δ	c	105	-10.77 (<0.001)	0	-10.88	0.263	$I(0)$
Romania, HICP, $100 \ln P$	c, t	260	-0.96 (0.949)	6	-0.88	0.314	$I(1)$
Romania, 12-month inflation π	c	260	-1.94 (0.315)	15	-2.31	0.404	inconclusive
its change $\Delta\pi$	c	259	-4.83 (<0.001)	14	-11.31	0.105	$I(0)$
US, real GDP, $100 \ln Y$	c, t	318	-1.54 (0.816)	1	-1.20	0.581	$I(1)$
its growth Δ	c	317	-15.51 (<0.001)	0	-15.44	0.456	$I(0)$
EUR/RON, $100 \ln S$	c	5 353	-1.22 (0.664)	7	-1.15	10.255	$I(1)$
daily returns	c	5 352	-27.42 (<0.001)	6	-60.82	0.055	$I(0)$
BET, $100 \ln P$	c, t	6 671	-1.90 (0.653)	19	-2.35	1.143	$I(1)$
daily returns	c	6 670	-16.50 (<0.001)	18	-74.55	0.360	$I(0)$
S&P 500, $100 \ln P$	c, t	6 715	-2.08 (0.557)	34	-2.36	2.281	$I(1)$
daily returns	c	6 714	-14.26 (<0.001)	35	-91.41	0.449	$I(0)$
Nile, yearly flow	c	100	-4.05 (0.001)	1	-6.38	0.869	conflict

- det.: deterministic terms (c: constant; t: trend); k: ADF lags (AIC); 5% critical values: ADF and PP about -2.86 (c), -3.41 (c, t); KPSS 0.463 (c), 0.146 (c, t)

TSA_ch3_unit_root_tests

Interpreting the unit-root table

- Log prices, the exchange rate, log GDP and log HICP: $I(1)$; their differences: $I(0)$
  - Romanian GDP: ADF p $= 0.43$ and KPSS $0.198 > 0.146$: a unit root, as Nelson and Plosser (1982) found for the US
- Romanian 12-month inflation: neither test rejects (KPSS $0.404 < 0.463$): inconclusive
  - a very persistent series, possibly with breaks; Sections 5 and 7
- Nile: both reject: a conflict
  - Chapter 1 showed a level shift in 1898: the next section explains the conflict
- Daily returns: PP gives far more negative values than ADF (BET: -74.6 against -16.5), but both reject; the verdict is the same

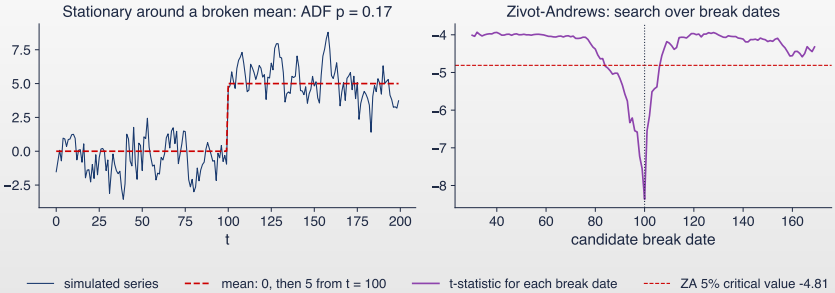
Recap: Phillips–Perron and KPSS

- PP: DF regression plus a long-run-variance correction; same H_0 and critical values as ADF
- KPSS: H_0 stationarity; large partial sums reject it
- Read ADF and KPSS together: agree, inconclusive, or conflict

Perron (1989): a break can look like a unit root

- Perron (1989): a stationary series around a mean or a trend that **shifts once** looks persistent
  - the DF regression explains the shift by $\hat{\phi}$ close to 1; the test loses power
- He re-examined Nelson and Plosser with two known breaks: the 1929 crash (level) and the 1973 oil shock (slope)
  - with dummy variables for the break, the unit root was rejected for most series
  - reading: most shocks are transitory; only a few rare events are permanent
- The critique: the break date was chosen after looking at the data

A level shift misleads the ADF test



- Left: AR(1) noise ($\phi = 0.6$) around a mean that jumps from 0 to 5 at $t = 100$; right: the t-statistic on y_{t-1} for every candidate break date, with a level dummy

TSA_ch3_structural_breaks

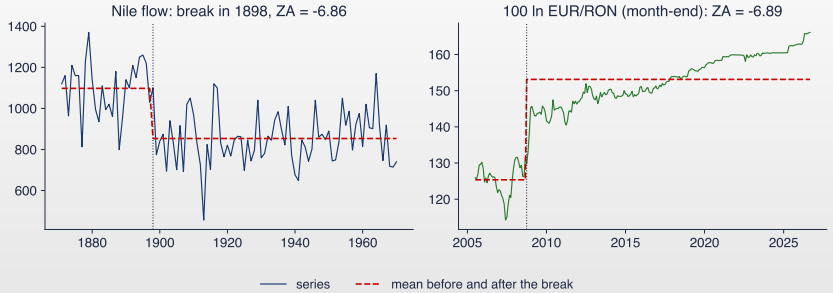
Interpreting the simulated break

- ADF with a constant: p = 0.17: the stationary series "has a unit root"
  - over 2 000 such series, the DF test rejects in 72% of cases; without the break: 100%
- With the break modelled, the Zivot–Andrews statistic falls to -7.47, far below the 5% value -4.81
  - the curve on the right reaches its minimum, -8.37, at the true date $t = 100$

Zivot and Andrews (1992): an unknown break date

- Zivot and Andrews (1992): let the data choose the break date T_B
  - for each T_B in the central 70% of the sample, run the ADF regression with a break dummy:
  - $\Delta y_t = c + \theta DU_t + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$, $DU_t = 1$ for $t > T_B$
  - DU_t: a dummy variable, 0 before and 1 after the break; θ: the size of the level shift
  - the statistic is the **minimum** t-ratio of γ over all dates
- Three versions: break in the level, in the slope, in both
  - critical values more negative than ADF: about -4.81 (level) and -5.07 (both) at 5%
- H_0: unit root without a break; H_1: stationary with one break
  - several breaks: Bai and Perron (1998); a rejection does not prove that the break is the right story

Zivot–Andrews on two real series



- Left: the Nile flow (statsmodels data set, 1871–1970); right: 100 ln EUR/RON, month-end BNR reference rate
- Dotted line: the break date chosen by the test (break in the level)

TSA_ch3_structural_breaks

Interpreting the two breaks

- Nile: ZA $= -6.86$, break in 1898, the mean falls from 1098 to 853
  - the date matches the history: the first Aswan dam (built 1898–1902) and a drier climate; the ADF–KPSS conflict is explained
- EUR/RON (month-end): ADF $p = 0.55$, but ZA $= -6.89$, break in October 2008: the 2008 depreciation (mean 3.50 lei before, 4.62 after)
  - a managed exchange rate: long calm periods and rare adjustments, not a pure random walk
  - but after 2008 the rate still drifts up: "stationary around one break" is also a simplification

Recap: Structural breaks

- A one-time shift in the mean or trend biases unit-root tests towards non-rejection
- Zivot–Andrews searches the break date; its critical values are more negative
- Unit root or break: check the dates against known events

The ARIMA model

- **Definition**: $y_t \sim$ ARIMA(p, d, q) (autoregressive integrated moving average) if $\Delta^d y_t$ is a stationary, invertible ARMA(p, q):
  - $\phi(L)\,(1 - L)^d\, y_t = c + \theta(L)\,\varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, \sigma^2)$
  - p: AR order; d: number of differences (unit roots); q: MA order; $\phi(L)$, $\theta(L)$ as in Chapter 2
- The AR polynomial of the levels is $\phi(z)(1 - z)^d$: d roots exactly on the unit circle
  - "integrated": y_t is a d-fold cumulative sum of an ARMA process
- Box, Jenkins and Reinsel (2008) made the model popular; in practice $d \leq 2$, and $d = 1$ most often

Special cases and the constant

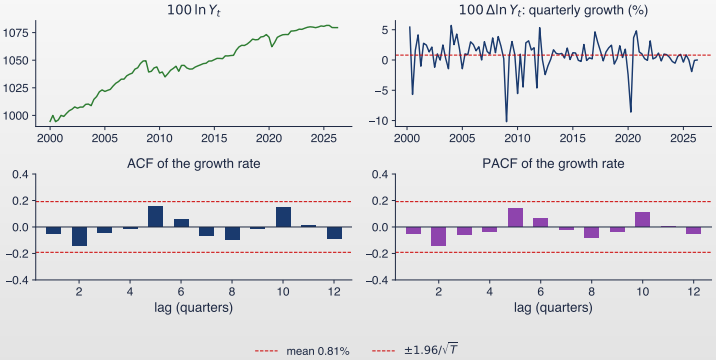
Model	Equation and meaning
ARIMA(0,1,0)	$\Delta y_t = c + \varepsilon_t$: random walk (with drift c)
ARIMA(1,1,0)	$\Delta y_t = c + \phi \Delta y_{t-\mathbf{1}} + \varepsilon_t$: growth with momentum
ARIMA(0,1,1)	$\Delta y_t = \varepsilon_t + \theta \varepsilon_{t-\mathbf{1}}$: simple exponential smoothing with $\alpha = 1 + \theta$ (Chapter 0)
ARIMA(0,2,2)	$\Delta^{\mathbf{2}} y_t = \varepsilon_t + \theta_{\mathbf{1}} \varepsilon_{t-\mathbf{1}} + \theta_{\mathbf{2}} \varepsilon_{t-\mathbf{2}}$: the model behind Holt's linear method

- The constant c changes the long-run forecast Hyndman and Athanasopoulos (2021), Ch. 9.7
  - $d = 0$: forecasts go to the mean; $d = 1$: $c \neq 0$ gives a straight line (drift), $c = 0$ a flat line
  - $d = 2$: $c \neq 0$ gives a quadratic trend, rarely sensible; use $c = 0$

The Box–Jenkins cycle with unit roots

- **1. Transform**: logarithm or Box–Cox if the variance grows with the level (Chapter 1)
- **2. Choose** d: plot, ACF, ADF and KPSS; test Δy_t first; stop at the first stationary difference
- **3. Choose** p, q: ACF and PACF of $\Delta^d y_t$ (Chapter 2), then compare candidates by AICc on the same d
- **4. Estimate** by maximum likelihood; decide on the constant
- **5. Check** the residuals: ACF, Ljung–Box with $m - p - q$ degrees of freedom, outliers
- **6. Forecast** the levels with intervals; go back to step 2 or 3 if the checks fail

Romanian GDP: identification



- Real GDP, seasonally adjusted (Eurostat), 2000Q1–2026Q2, $T = 106$; top: $100 \ln Y_t$ and the quarterly growth; bottom: ACF and PACF of the growth

TSA_ch3_arima_gdp

Interpreting the identification plots

- $d = 1$: the level is $I(1)$ (Section 4); the growth rate is stationary, mean 0.81% per quarter (about 3.2% per year)
  - two large falls: -10.2% in 2009Q1 (global financial crisis) and -8.6% in 2020Q2 (pandemic)
- p and q: no ACF or PACF value outside ± 0.19; $\hat{\rho}(1) = -0.05$, $\hat{\rho}(2) = -0.14$
  - Ljung–Box $Q^*(8) = 7.3$ (p = 0.51): the growth rate looks like white noise around its mean
- The plots suggest ARIMA(0,1,0) with drift; the next slide checks the alternatives

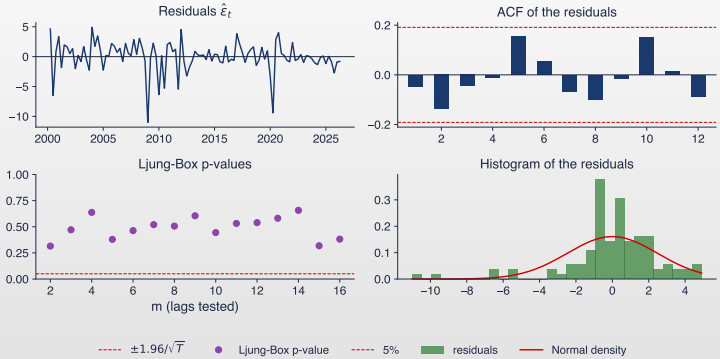
Model choice by AICc

	$q = 0$	$q = 1$	$q = 2$
$p = 0$	491.9	493.7	493.6
$p = 1$	493.8	494.9	495.8
$p = 2$	493.7	495.7	495.4†

- ARIMA$(p, 1, q)$ with drift for $100 \ln Y_t$, estimated by maximum likelihood; AICc: AIC with the small-sample correction (Chapter 2); †: the optimiser did not converge
- Lowest AICc: ARIMA(0,1,0); the next model is 1.7 points worse
  - differences below 2 points are not meaningful: keep the simplest model
- Estimate: drift $\hat{c} = 0.81$ (SE 0.30, $t = 2.72$), $\hat{\sigma} = 2.47$: $100\,\Delta \ln Y_t = 0.81 + \hat{\varepsilon}_t$

TSA_ch3_arima_gdp

Romanian GDP: residual checks



- Residuals of ARIMA(0,1,0) with drift: time plot, ACF, Ljung–Box p-values for $m = 2, \ldots, 16$, histogram with a Normal density

TSA_ch3_arima_gdp

Interpreting the residual checks

- No autocorrelation left: $Q^*(8) = 7.3$ (p $= 0.51$), $Q^*(12)$ p $= 0.54$; all p-values above 5%
- Not Normal: kurtosis 7.9, Jarque–Bera 145; the largest residual is -11.0 in 2009Q1
  - two crises dominate the tails; the Normal intervals of the next slides are too narrow in a crisis and slightly too wide in calm times
  - options: dummy variables for 2009 and 2020, or a bootstrap of the residuals for the intervals

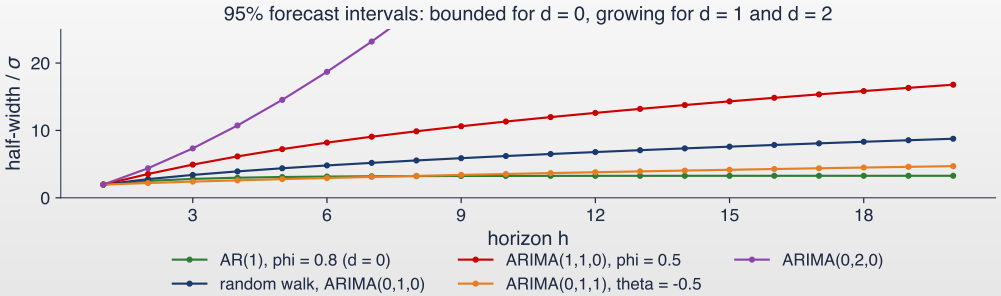
Forecasting with ARIMA

- **Point forecasts**: write the model for the levels and iterate it, with $\hat{\varepsilon}_{T+j} = 0$ for $j \geq 1$
  - ARIMA(1,1,0): $y_t = (1+\phi)y_{t-1} - \phi y_{t-2} + \varepsilon_t$
- **Forecast error**: $y_{T+h} - \hat{y}_{T+h} = \sum_{j=0}^{h-1} \psi_j \varepsilon_{T+h-j}$, with $\psi(z) = \theta(z)/[\phi(z)(1-z)^d]$
  - variance $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$; 95% interval $\hat{y}_{T+h} \pm 1.96\,\sigma\sqrt{\sum_{j<h} \psi_j^2}$
  - for $d \geq 1$ the ψ_j do not decay to 0: the variance grows without bound
- Two closed forms
  - random walk: $\psi_j = 1$, variance $h\sigma^2$: the interval grows like $\sqrt{h}$
  - ARIMA(0,1,1): $\psi_j = 1 + \theta$ for $j \geq 1$, variance $\sigma^2[1 + (h-1)(1+\theta)^2]$

Worked example: ARIMA(1,1,0) forecasts

- $\Delta y_t = 0.6\,\Delta y_{t-1} + \varepsilon_t$, $\sigma = 2$; last data: $y_T = 108$, $\Delta y_T = 5$
- **Point forecasts**: forecast the change, then add it
  - $\widehat{\Delta y}_{T+1} = 0.6 \cdot 5 = 3$, $\hat{y}_{T+1} = 111.00$; $\widehat{\Delta y}_{T+2} = 1.8$, $\hat{y}_{T+2} = 112.80$; $\hat{y}_{T+3} = 113.88$
- ψ **weights**: $\psi_0 = 1$, $\psi_1 = 1 + \phi = 1.60$, $\psi_2 = 1 + \phi + \phi^2 = 1.96$
  - variances: 4, $4(1 + 1.60^2) = 14.24$, $4(1 + 1.60^2 + 1.96^2) = 29.61$
  - 95% half-widths: 3.92, 7.40, 10.66: the interval almost triples in three steps
- The momentum $\phi = 0.6$ makes the uncertainty grow faster than for a random walk ($\sqrt{3} = 1.73$ times in three steps)

How fast the intervals grow



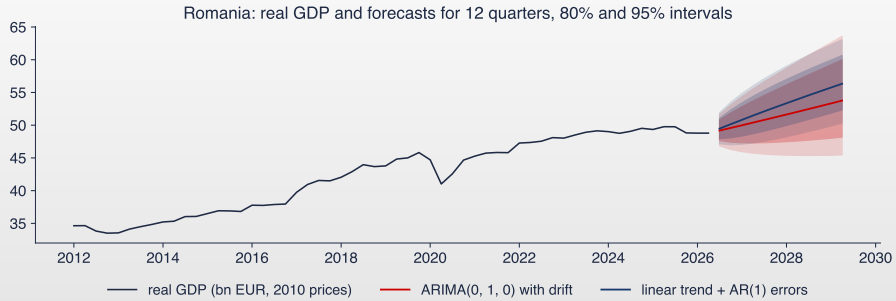
⊡ Half-width of the 95% forecast interval, in units of σ, for horizons 1 to 20, from the ψ weights of five models

TSA_ch3_forecast_intervals

Interpreting the interval widths

- $d = 0$, AR(1) with $\phi = 0.8$: the width levels off at $1.96\sigma/\sqrt{1-\phi^2}$ (3.27σ at $h = 20$)
- $d = 1$: growth like $\sqrt{h}$; at $h = 20$: 8.77σ (random walk), 16.78σ (ARIMA(1,1,0), $\phi = 0.5$), 4.70σ (ARIMA(0,1,1), $\theta = -0.5$)
  - a negative MA part offsets part of each shock: narrower intervals
- $d = 2$: growth like $h^{3/2}$ (105.00σ at $h = 20$): over-differencing makes long-run forecasts useless

Romanian GDP: two forecasts



- ARIMA(0,1,0) with drift (red) and a linear trend with AR(1) errors (blue), both estimated on 2000Q1–2026Q2; forecasts to 2029Q2, transformed back to bn EUR (2010 prices)

TSA_ch3_arima_gdp

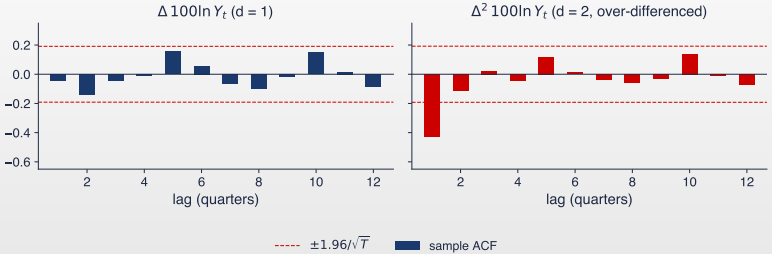
Interpreting the GDP forecasts

- The random walk with drift grows by 0.81% per quarter from the last value (48.8 bn EUR, 2010 prices, in 2026Q2): 53.8 bn after 12 quarters
  - 95% half-width (in % of GDP): 4.8 after 1 quarter, 9.7 after 4, 16.8 after 12, i.e. $\times 3.46 = \sqrt{12}$
- The TS model pulls GDP back towards its old line ($\hat{\phi} = 0.92$): 56.3 bn after 12 quarters, half-width 11.2%
  - more optimistic and more confident, because it assumes the recent slowdown is temporary
- The tests favour DS at 5% (ADF p = 0.428; KPSS 0.198 $>$ 0.146), but are inconclusive at 1% (KPSS $<$ 0.216): the choice of d is a forecasting decision with consequences

Over-differencing

- Differencing an $I(0)$ series creates an MA unit root: $\Delta\varepsilon_t = \varepsilon_t - \varepsilon_{t-1}$, $\theta = -1$ (Chapter 1)
  - the result is not invertible: no AR(∞) representation, poor estimation Plosser and Schwert (1977)
- Symptoms
  - $\hat{\rho}(1)$ of the differenced series close to -0.5
  - the variance increases after differencing instead of decreasing
  - an estimated MA coefficient close to -1
- Rule: difference only as long as the variance goes down and the tests ask for it

Over-differencing Romanian GDP



- ACF of the quarterly growth $\Delta\, 100 \ln Y_t$ $(d = 1)$ and of its difference $\Delta^2\, 100 \ln Y_t$ $(d = 2)$

TSA_ch3_arima_gdp

Interpreting over-differencing

- $d = 1$: $\hat{\rho}(1) = -0.05$, variance 6.16; $d = 2$: $\hat{\rho}(1) = -0.43$, variance 12.80: all three symptoms
- An MA(1) fitted to $\Delta^2 100 \ln Y_t$ gives $\hat{\theta} = -0.987$ (SE 0.080): practically -1, the extra difference is cancelled by the MA part
- Correct choice: $d = 1$

Recap: ARIMA models

- ARIMA(p, d, q): an ARMA(p, q) for $\Delta^d y_t$; the constant decides the long-run slope
- Romanian GDP since 2000: ARIMA(0,1,0) with drift, 0.81% per quarter; residuals uncorrelated but heavy-tailed
- Forecast variance $\sigma^2 \sum_{j<h} \psi_j^2$: bounded for $d = 0$, growing for $d \geq 1$
- Over-differencing: $\hat{\rho}(1) \approx -0.5$, larger variance, MA coefficient near -1

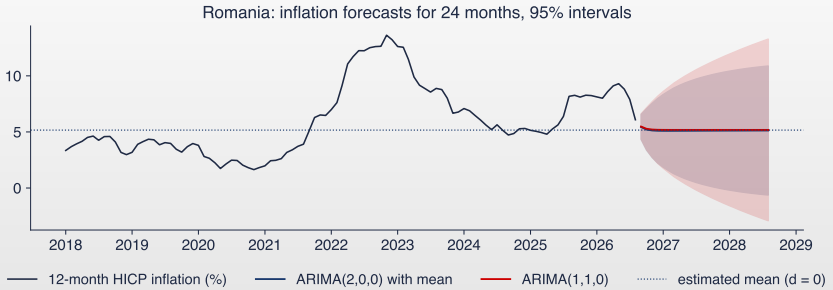
Automatic ARIMA

- Hyndman and Khandakar (2008) (`auto.arima` in R; `pmdarima` and `statsforecast` in Python):
  - d: repeated KPSS tests at 5%: difference while KPSS rejects
  - p, q: a stepwise search from a few starting models, by AICc, on the chosen d; a constant or drift if $d \leq 1$
  - rejects models with roots too close to the unit circle and fits that fail
- Useful for many series at once and as a starting point
  - evaluated by its authors on the 3 003 series of the M3 forecasting competition, it was competitive with the best methods there
- But each step embeds a choice that you must be able to defend (next slide)

Automatic ARIMA: checks

- **AICc is not comparable across** d: with $d = 1$ the likelihood is computed for Δy_t, a different series
  - choose d by tests and judgement, then compare p, q
- **Near-cancelling roots**: Romanian GDP from 1995Q1: the lowest AICc, 607.5 (ARIMA(0,1,0): 611.1), belongs to ARIMA(2,1,2), whose fit did not converge
  - MA roots of modulus 1.0002, AR roots 1.014: AR and MA almost cancel; $\hat{\theta}_2 = 1.000$ with SE 6.96
  - a model fitted to the noise of the 1990s data, not a description of the economy
- **Always**: plot the data, check the convergence and the roots, run the residual tests, compare with a simple benchmark (random walk, Chapter 0)

Romanian inflation: $d = 0$ or $d = 1$?



- 12-month HICP inflation (Eurostat), January 2005–August 2026; best model by AICc for $d = 0$ (with a mean) and for $d = 1$, $p \leq 3$, $q \leq 2$; forecasts for 24 months

TSA_ch3_arima_in_practice

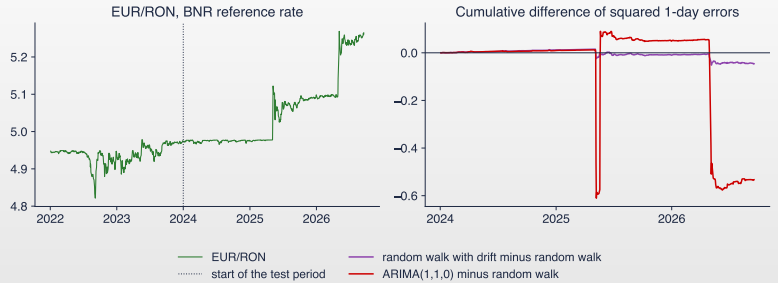
Interpreting the two inflation models

- $d = 0$: ARIMA(2,0,0), $\hat{\phi}_1 = 1.32$, $\hat{\phi}_2 = -0.34$, sum 0.976: stationary, but barely; mean 5.2%
  - $d = 1$: ARIMA(1,1,0), $\hat{\phi} = 0.33$: almost the same short-run dynamics
- After 24 months: 5.1% in $[-0.6, 10.9]$ against 5.2% in $[-2.9, 13.3]$
  - similar points, different intervals: d decides how much uncertainty we report
- KPSS (0.404) keeps $d = 0$, as automatic ARIMA would; ADF ($p = 0.32$) suggests $d = 1$
  - both models leave autocorrelated residuals: 12-month rates overlap and inherit the season; Chapter 4 models monthly inflation

Case study: Meese and Rogoff (1983)

- **Question**: can exchange-rate models forecast better than a random walk?
  - Meese and Rogoff (1983): monetary models of the 1970s, time series models (ARIMA, VAR) and the random walk; dollar rates, horizons of 1 to 12 months
- **Finding**: out of sample, none beat the random walk in RMSE (root mean squared error)
  - even with the actual future values of the fundamentals plugged in
- **Legacy**: the "Meese–Rogoff puzzle" and the random walk as the benchmark for every exchange-rate forecast
  - next chart: the same comparison for EUR/RON

EUR/RON: ARIMA against the random walk



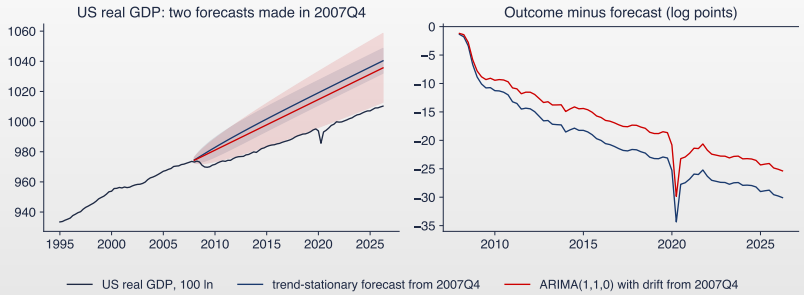
- Parameters estimated on 4 673 daily changes until December 2023; 679 one-day-ahead forecasts from January 2024; right: the cumulated difference of squared errors (below 0: the model beats the random walk)

TSA_ch3_arima_in_practice

Interpreting the EUR/RON comparison

- RMSE of the daily change (100 ln): random walk 0.1330, with drift 0.1327, ARIMA(1,1,0) 0.1300 ($\hat{\phi} = 0.169$)
  - a gain of 2.2%; over 20 days (33 non-overlapping windows): 0.667 against 0.638, a gain of 4.3%
- The gain comes from a few days: two jumps of the rate (from 4.9746 to 5.2644 lei over the period) continued on the next day
  - on calm days the random walk is as good; a formal comparison needs the Diebold–Mariano test (Chapter 4)
- Lesson of Meese and Rogoff: report the random walk next to any exchange-rate forecast

A classic series: US real GDP after 2007



- FRED series GDPC1 since 1947; a TS model (linear trend, AR(2) errors) and a DS model (ARIMA(1,1,0) with drift) estimated up to 2007Q4 and forecast to 2026Q2; 95% intervals

TSA_ch3_us_gdp

Interpreting the US GDP forecasts

- The TS model expected a return to the pre-2008 line; the gap kept widening: -30 log points at the end (half-width of its interval: 8)
  - the 2008–2009 loss was never recovered: evidence for a permanent shock, as the DS view of Nelson and Plosser (1982) implies
- The DS model also misses (-25), but its interval (half-width 23) at least warns of large uncertainty
  - both assume the same drift as before 2008; growth after 2008 was slower: a break in the drift (Section 5)
- Unit-root tests on the full series: ADF p $= 0.82$, KPSS 0.58: $I(1)$

Recap: ARIMA in practice

- Automatic ARIMA: KPSS for d, AICc for p, q; check convergence, roots and residuals
- AICc compares models with the same d only
- Exchange rates: the random walk is a hard benchmark Meese and Rogoff (1983)
- TS or DS changes the long-run forecast and its interval: US GDP after 2008

Possible contribution of AI

- **Code**: a first draft of a script that runs ADF, PP and KPSS on many series and tabulates the verdicts
- **Explanation**: a second explanation of why the Dickey–Fuller distribution is not Normal, or of a ZA output
- **Exploration**: unit roots and breaks in all EU inflation rates, or ARIMA benchmarks for many Romanian series
- Example prompt
  - `Write Python code that loads Romanian quarterly real GDP (Eurostat namq_10_gdp, seasonally adjusted), runs ADF with constant and trend and KPSS on 100*log GDP and on its difference, fits ARIMA(p,1,q) with drift for p, q <= 2, and plots 8-quarter forecasts with 95% intervals.`

Checks you must run

- The direction of each test: ADF and PP have the unit root as H_0, KPSS has stationarity as H_0
- The deterministic terms, the number of lags and the critical values actually used (Dickey–Fuller, not Normal)
- That "not rejected" is not reported as "proved"
- That regressions between trending series are not presented as causal evidence
- Convergence warnings, roots near the unit circle, residual tests, and AICc compared only for the same d
- Every cited reference: it must exist; check the DOI

Key takeaways

- Trending series are TS (shocks fade) or DS (shocks are permanent); the treatment differs: detrend or difference
- Regressions between unrelated $I(1)$ series are spurious: high t and R^2, DW near 0
- ADF and PP test H_0: unit root with Dickey–Fuller critical values; KPSS tests H_0: stationarity; use them together
- The tests have low power near $\phi = 1$ and are misled by breaks; Zivot–Andrews allows one break
- ARIMA(p, d, q) = ARMA on $\Delta^d y_t$; its forecast intervals grow with the horizon when $d \geq 1$
- Automatic selection is a starting point: check d, convergence, roots, residuals and a random-walk benchmark

Key formulas

Quantity	Formula
TS and DS	$y_t = \alpha + \beta t + u_t, \quad \Delta y_t = \beta + u_t, \quad u_t \sim I(0)$
ADF	$\Delta y_t = c + bt + \gamma y_{t-\mathbf{1}} + \sum_{j=\mathbf{1}}^{k} \delta_j \Delta y_{t-j} + \varepsilon_t, \quad H_{\mathbf{0}}\colon \gamma = 0, \quad \tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$
Lags	$k_{\max} = 12\,(T/100)^{\mathbf{1/4}}$, then AIC or BIC
KPSS	$\eta = \sum_{t=\mathbf{1}}^{T} S_t^{\mathbf{2}}/(T^{\mathbf{2}}\hat{\lambda}^{\mathbf{2}}), \quad S_t = \sum_{s \le t} e_s, \quad$ 5%: 0.463 (c), 0.146 (c, t)
ARIMA(p, d, q)	$\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$
Forecast variance	$\sigma^{\mathbf{2}} \sum_{j=\mathbf{0}}^{h-\mathbf{1}} \psi_j^{\mathbf{2}}, \quad \psi(z) = \theta(z)/[\phi(z)(1-z)^d]$; random walk: $h\sigma^{\mathbf{2}}$
ARIMA(0,1,1)	$\sigma^{\mathbf{2}}[1 + (h-1)(1+\theta)^{\mathbf{2}}], \quad \alpha_{\mathbf{SES}} = 1 + \theta$

Self-assessment

- **Question**: an ADF test with constant and trend gives $\tau = -2.9$ with $T = 200$. Do you reject the unit root at 5%?
  - **Answer**: no: $-2.9 > -3.43$; with the Normal table you would wrongly reject
- **Question**: ADF does not reject and KPSS does not reject. What do you conclude?
  - **Answer**: inconclusive: the sample is too short to decide; report both models, or choose d by the purpose
- **Question**: for ARIMA(0,1,1) with $\theta = -0.6$ and $\sigma = 1$, what is the variance of the 5-step forecast error?
  - **Answer**: $1 + 4 \cdot 0.4^2 = 1.64$
- Next: Chapter 4, seasonality: seasonal differences, SARIMA, TBATS and Prophet

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